

A Filterless Design For AIR-POLLUTION CONTROL



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This article takes a look at the causes, effects and control of air pollution in the urban environment. It also introduces easy-to-construct home and commercial air-purifier systems to effectively control air pollution, and release clean and safe-to-breathe air back into the environment.

Air pollution contains one or more contaminants like dust, harmful gases, smoke, pollen and other allergens.

Normal composition of air is:

Nitrogen	- 78 %
Oxygen	- 21 %
Argon	- 0.9 %

Carbon-dioxide	- 0.037 %
Ozone, helium and ammonia	- Traces
Water vapour	- Balance

Table I indicates the mix of pollutant gases released from petrol and diesel based vehicles.

Table II summarises the source of major pollutants and their impact on the health of human beings.

We now introduce you to system designs for easy-to-construct air purifiers for home and commercial uses. (Note: A prototype of the air purifier for home use has already been tested successfully by the author.)

Air-purifier system for home and industrial use

Commercially, a large number of air purifiers are available for domestic use, but are quite expensive. You can make a simple air purifier for domestic use using easily available materials from the market. The same can be customised for use in hazardous industries as well, to improve the working conditions and health of the

TABLE I
POLLUTANTS FROM VARIOUS VEHICLES

Gases	Carbon monoxide	Hydrocarbons	Oxides of nitrogen	Particulate matter	Carbon-dioxide
Petrol car without catalytic converter	100	100	100	—	100
Petrol car with catalytic converter	42	19	23	—	100
Diesel car without catalytic converter	2	3	31	100	85

*100 is taken as reference value for comparison

TABLE II
POLLUTANT SOURCES AND HEALTH EFFECTS

Source	Causes/sources	Health effects
Carbon monoxide. It is a colourless, odourless and tasteless gas that is poisonous to hemoglobin animals including human beings. It is formed by incomplete combustion of carbon-containing fuels.	Sources of carbon monoxide include cigarette smoke and incomplete combustion of fossil fuels (more than 77 per cent comes from motor vehicle exhaust).	Reduced ability of red blood cells to carry oxygen to body cells and tissues, which leads to headaches and anaemia. At high levels, it can cause coma, irreversible brain damage and even death.
Nitrogen dioxide. It is a reddish-brown gas that has a characteristic sharp biting odour and is a prominent air pollutant in the atmosphere. It gets converted into nitric acid (HNO ₃) when it comes in contact with water vapours.	It is caused by burning fossil fuels in industries and power plants.	Lung irritation and damage. Environmental effects involve acid deposition leading to damage of trees, lakes, soil and ancient monuments.
Sulphur dioxide. It is a toxic gas with a pungent, irritating smell.	It is formed by combustion of sulphur-containing fossil fuels such as coal and oil. In the atmosphere it is converted into sulphuric acid.	Irritates the nose, throat and airways, and causes coughing, wheezing, shortening of breath and tightening around the chest.
Photochemical smog. It is a brownish smoke that frequently forms on clear, sunny days over large cities with significant amounts of automobile traffic.	It is caused mainly due to chemical reactions among nitrogen oxides and hydrocarbons in the presence of sunlight.	Breathing problems, cough, eye, nose and throat irritation, heart diseases, reduced resistance to colds and pneumonia. Environmental effects involve damage to plants and trees. Additionally, smog reduces visibility.
Suspended particulate matter. This contains a variety of particles and droplets (aerosols) that can be suspended in the atmosphere for short to long periods of time.	These are produced by burning of coal in power plants and industries. Also produced by diesel vehicles.	Nose and throat irritation, lung damage, bronchitis, asthma, reproductive problems and cancer.